CODEMIGRATE · PROJECT REVIEW

# Demo script (about 4 minutes)

Presenter: **M5** (web app). **M4** speaks during step 6 (validator).

> Use the **offline** reasoner for the live part: no API key, no quota, and the same result every time. Say the lines in your own words; do not read them out.

## Before the review (30 minutes early)

1. Start the API (terminal 1, in `D:\CodeMigrate`):
   `.venv\Scripts\python.exe -m uvicorn codemigrate.api.main:app --port 8000`
2. Start the UI (terminal 2): `cd frontend; npm run dev`
3. Open `http://localhost:5173`, log in once with the demo account, keep the tab open.
4. Open `docs/demo/views.py` in an editor so you can copy it quickly.
5. Do one full practice run of the steps below. Close other heavy apps.
6. Keep the backup ready: a screen recording of this demo, plus `docs/CodeMigrate_Review.pptx`.

## The demo input: docs/demo/views.py

```
import datetime

from django.db import models
from django.http import JsonResponse
from django.utils.timezone import utc


class Order(models.Model):
    customer = models.CharField(max_length=100)
    created = models.DateTimeField()

    class Meta:
        index_together = [["customer", "created"]]


def recent_orders(request):
    now = datetime.datetime.now(tz=utc)
    orders = Order.objects.filter(created__lte=now)
    return JsonResponse({"count": orders.count()})
```

It looks normal and works on Django 4.0, but has two problems for 5.1: `django.utils.timezone.utc` (deprecated 4.1, removed 5.0) and `Meta.index_together` (deprecated 4.2, removed 5.1).

## Steps

| # | Do | Say (in your own words) |
|---|---|---|
| 1 | Show the **New scan** page. | "This is a small Django view that runs fine on Django 4.0. We want to upgrade it to 5.1." |
| 2 | Project: type `Demo shop`. From **4.0**, To **5.1**, Reasoner **offline**, Retriever **hybrid**. Tab **Paste code**, File path `views.py`, paste the code, click **Scan**. | "We choose the versions. Offline mode needs no API key; with an LLM the steps are the same." |
| 3 | Wait for **COMPLETED** (about 10 s). Point at the **3 findings**: lines 5 and 17 (`utc`), line 13 (`index_together`). | "Every finding points to an exact line and is VERIFIED, which means our validator checked its citation." |

| # | Do | Say (in your own words) |
|---|---|---|
| 4 | Click the release-note link on the utc finding (opens docs.djangoproject.com, 4.1 notes). Come back. | "Each warning links to the official Django release note. Nothing is shown without a source." |
| 5 | Tab **Updated code**: show `utc = datetime.timezone.utc` replacing the old import and the `# TODO(codemigrate)` above index_together. Then tab **Changes** (the diff). Click **Download views.py**. | "Safe changes are fixed automatically. Where an automatic fix could be wrong, we add a clear TODO instead of guessing." |
| 6 | **M4** runs the command below in a terminal (prepared beforehand). | "This is a real answer from Gemini on our benchmark. It claimed `Meta.unique_together` breaks but gave no release-note citation. The validator rejected it as UNSUPPORTED_CLAIM, so the user never sees it. That is how we reach 0% made-up findings." |
| 7 | Back in the browser: **Export PDF** (shows the report), then **Compare systems** (results table). | "Reports can be shared. Here are the benchmark results: F1 0.82 versus 0.02 for django-upgrade." |

## Command for step 6 (one line, PowerShell)

```
Get-Content docs\demo\gemini_rejected_claims.jsonl | ConvertFrom-Json |
  Where-Object status -eq 'UNSUPPORTED_CLAIM' |
  Format-List file, line, symbol, status, reject_reason
```

Type it as one line. Expected output:

```
file          : djmoney/contrib/exchange/models.py
line          : 30
symbol        : Meta.unique_together
status        : UNSUPPORTED_CLAIM
reject_reason : no citation
```

`docs/demo/gemini_rejected_claims.jsonl` is a copy of every claim the validator rejected in the Gemini test run.

## Expected Updated code (step 5)

```
import datetime

from django.db import models
from django.http import JsonResponse
utc = datetime.timezone.utc


class Order(models.Model):
    customer = models.CharField(max_length=100)
    created = models.DateTimeField()

    class Meta:
        # TODO(codemigrate): index_together: deprecated in Django 4.2. Use Meta.indexes.
        index_together = [["customer", "created"]]
...
```

## Optional: command line (30 seconds, if time allows)

```
.venv\Scripts\codemigrate.exe scan docs\demo\views.py --provider offline --write-fixed fixed_out
```

Prints the same 3 findings and writes the updated file to `fixed_out\views.py`; the original is never changed.

## If something goes wrong

| Problem | Do this |
|---|---|
| Page does not load | Check both terminals are running; reload `http://localhost:5173`. |
| Scan stays RUNNING for more than 30 s | The first scan loads the retrieval model; wait once more, otherwise switch to the backup video. |
| Login fails | Click Register and create a new account (takes 10 s). |
| Anything else | Say “let me show the recorded run”, play the backup video, continue with the slides. |

## Questions you may get during the demo

**Q: Why offline and not the LLM?**

A: Same pipeline and same validator. Offline is repeatable and does not depend on internet or API quota. Offline F1 is 0.81; with Gemini it is 0.82.

**Q: Why didn't it fix index_together automatically?**

A: Moving to Meta.indexes needs a migration and naming choices. A wrong automatic fix is worse than a clear TODO.

**Q: Does the validator ever reject correct findings?**

A: Yes, sometimes: in the benchmark the LLM gave a few correct warnings with the wrong symbol name, and they were dropped. That is why recall goes from 0.92 (C) to 0.88 (D). In exchange, no made-up findings get through.